m_t/m_b = C_2·g = 42 — adjudication (HOLD at Identified), and it is a live test case for the quark-generation boundary-distance lead

Keeper

2026-07-28

K965 — m_t/m_b adjudication

The question (from K963): m_t/m_b = C_2·g = 42 sits at IDENTIFIED, 1.7%. Promote to DERIVED (is T1990 "the b's home is the top" a *forcing route*) or hold (is it a *placement*)?

Ruling: HOLD at IDENTIFIED

Two reasons, and they agree:

1. T1990 is a placement, not a demonstrated forcing route. "The b's home is the top" locates b and t as a third-generation doublet; it does not exhibit a target-innocent mechanism that *forces the ratio value* to be C_2·g. 42 = 6·7 is a clean two-integer monomial that matches ~41.3, but a monomial match with a home-statement is the IDENTIFIED signature (single route, forcing open), not the DERIVED one. Under the open-piece test the residual (42 vs ~41.3, 1.7%) is value-bearing — nothing in the corpus pins it to sub-% — so it does not clear the bar. *(If Lyra later exhibits a forced constraint inside it — e.g. the doublet's boundary structure fixing the ratio up to a named residual — it becomes a Ceiling/Value candidate per K964, earned by the three conditions, not before.)*
2. ★ The 1.7% is EXPECTED under the quark-generation lead — the tier and the lead cross-validate. m_t/m_b is a gen-3 / gen-3 ratio: both quarks are the most boundary-localized (F603/K768 mass-ordering = boundary-reach). By the boundary-distance lead (team prompt 2026-07-28b), values read near the Shilov boundary are *limits, not pinned bulk cells*, so they are inherently coarser — exactly the observed drift (m_u/m_d 0.09% → m_t/m_b 1.7%). So m_t/m_b being 1.7%-coarse is not a defect to explain away; it is a data point for the lead. Holding it at Identified is consistent with "this value is supposed to be a boundary-limit, not a clean cell."

Consequence — a prediction the lead now owns

This makes m_t/m_b a live test case: when Lyra computes the radial discrete-series addresses, the quark-generation theory should predict the m_t/m_b precision (~1.7%) from the two quarks' boundary-distances — not just tolerate it. If the predicted blur matches the observed 1.7% (and the analogous m_c/m_s, V_cb blurs), the lead earns promotion from Insight-LEAD toward a real theory, and m_t/m_b retro-confirms it. If the predicted blur does NOT track, the lead is weakened — m_t/m_b is one of the cleaner falsification points because its coarseness is mid-range, not extreme.

Net: m_t/m_b stays IDENTIFIED; it is not a soft-Derived candidate; and it is now a named test/falsification point for the quark-generation boundary-distance lead. Cal's soft-promotion skeptic pass (m_u/m_d, λ_H) does not extend to this row.

— K965, Keeper, 2026-07-28. Resolves the last open K963 adjudication; links the tier ruling to the quark-generation lead as mutual corroboration. See [[Keeper_K963_BOTTOM_UP_TIER_REVIEW_apply_sharpened_K962_open_piec
07-28]], [[Keeper_K964_CEILING_VALUE_DECOMPOSITION_of_the_Identified_tier_three_earning_conditions_2026-07-28]].